Topic: Design and implement a Smart monitoring and control system for greenhouse

Report submitted in fulfilment of the requirements for the Diploma

Diploma in Electrical Engineering

Department of Electrical Engineering

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Diploma in Electrical Engineering

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Date: 10/03/2025

Declaration

I Amahle declare that this report is my original work and that it has not been presented to any other university or institution for similar or any other degree award. It is being submitted for Engineering programming 3 (EIENP3A) to the Department of Electrical Engineering at the Vaal University of Technology, Vanderbijlpark.

A. Notununu 15 May 2025

Signature Date

Acknowledgment

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Dedication

This initiative honours the countless individuals who work tirelessly to build a sustainable future. To the engineers whose creativity and expertise lead to more clever solutions, and farmers who imagine a more ecologically friendly future.

As an engineering student, I dedicate this Smart monitoring and control system for greenhouse to my mentors, family, and friends from Qumbu. Your constant guidance, encouragement, and support, as well as the passion and hard work you put into farming to create high-quality food and livestock, have been the driving force behind this attempt.

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ACRONYM AND ABBREVIATION

LDR	Light-dependent Resistor
PIR	Passive Infrared
IR	Infrared
IDE	Integrated Development Environment
ESD	Electrostatic Discharge
IEC	International Electrotechnical Commission
FCC	Federal Communication Commission
MQTT	Message Queuing Telementry Transport
LoRa	Long Range

Chapter 1: Introduction and purpose of Smart monitoring and control system for greenhouse

1.1 Introduction

Among those are the agricultural engineers who need to maximize the output and quality of different crops by keeping greenhouse climatic conditions ideal. The importance of the food populace has been rising day by day due to the increase of the international population (Prof. Khaldun I.A & Hind F. A, 2015). One way to avoid hunger and fulfil people's food needs is to increase agricultural production. Research on these and other effects of these factors on plant growth and resource use efficiency is still important.

This project dives into how we can design and set up a smart greenhouse control system. It sits at the crossroads of sustainable farming and tech. This system is crucial as it helps to monitor and manage the greenhouse environment, making sure plants have the best conditions to grow and avoiding resource wastage. It’s innovative and practical. Focusing on one parameter at a time enhances farming practices. The Smart monitoring and control system for greenhouse will improve the agricultural environment.

1.2. Background

In the agricultural environment, blending automation, data analytics, and some sensor tech help to create the ideal environment for crops to thrive. The Smart monitoring and control system for greenhouse allows for accurate control of important parameters, such as ground steam, light size, temperature, and steam. Keeping the temperature and humidity at optimal levels doesn’t just help with crop growth; it also prevents diseases and pests, which is a big win for farmers looking to increase their yield and workforce efficiency (Kim, E, 2017). Usually, conventional farming is critical to the emergence of challenges such as increasing the number of natural resource waste signs, frequent weather patterns, and the negative impact of chemical fertilizers on the environment.

A study that's worth mentioning here is by Abou-Mehdi-Hassani et al. (2025). They dive deep into how these smart technologies are being utilized in greenhouse systems. It's clear from their findings just how vital IoT is for accurate monitoring and control. They sifted through 54 papers from the Scopus database, covering the years 2019 to 2023, and they identified four main clusters: precision agriculture, smart greenhouses, IoT itself, and a combined category that deals with sustainability and Industry 4.0. Their paper sheds light on what drives the push for smart farming tech to create agricultural systems that are not just more effective but also kinder to the environment. They offer some practical solutions and a solid base for future advancements.

In 2023, Zaguia carried out an interesting study about a smart greenhouse control system based on smart technology. With sensor indication and an adaptive PID controller, it intelligently regulates temperature and humidity. Features include a real-time data visualization cloud system and a remote control on a smartphone application. The technologies improve crop productivity, reduce energy consumption, and generally increase the management for a greenhouse.

When you put all these studies together, it highlights the significance of smart control systems in greenhouse agriculture. They not only optimize resource use but also ramp up crop production and back sustainable farming practices all by blending state-of-the-art sensor tech with automated controls. It’s an exciting time for agriculture!

(zagui)

Figure 1: Greenhouse

1.3 **Problem statement**

Traditional methods of running greenhouses cause problems like running out of resources, climate control being unpredictable and monitoring that just takes forever. All this can lead to lower crop yields, higher costs, and resource wastage. With automation, data analysis, and some fancy sensor tech, these systems have been developed to make greenhouse management a lot more sustainable and effective.

A Smart monitoring and control system for Greenhouse can help to monitor and control.key environmental metrics such as temperature, humidity, motion and light intensity. It also incorporates a web server to facilitate remote monitoring and utilizes a database to ensure scalable and secure data storage. All of this is done in real-time. This ensures that everything operates smoothly by optimizing the greenhouse's microclimate, which boosts crop yields and makes better use of resources.

1.4 **Significance of the study**

The way the smart monitoring and control system for greenhouse have been integrated into greenhouses has changed the agricultural environment. The better management of natural resources, greater attention to sustainability, and a significant increase in farm production. Through the integration of data analysis, automation tools, and modern cognitive technologies, it creates the optimal environment for plant growth. You can closely monitor and adjust critical parameters such as soil moisture, light intensity, temperature, and humidity as needed.

The main purpose of the Smart monitoring and control system for Greenhouse is to create a harmonious and automated environment by allowing the plant to weaken and perform natural resource management. It is important to collect and evaluate data from monitoring light levels, regulating temperature, detecting plant health problems, monitoring humidity levels, automating environmental changes, and providing remote monitoring. Utilizing a web server for remote monitoring alongside a database for secure and efficient data management, the framework equips farmers with actionable insights that are essential for making informed decisions and achieving long-term optimization. This solution not only enhances crop productivity but also encourages sustainable resource utilization and provides a scalable model that can be tailored to various agricultural requirements. By merging traditional farming techniques with contemporary Industry 4.0 technologies, it supports environmental preservation, economic resilience, and food security, thereby presenting a transformative method to address the challenges posed by an expanding global population while promoting precision agriculture

1.5 Objectives

The study's goals lay a clear path and some attainable milestones for rolling out a Smart monitoring and control system for greenhouse. They’re designed to be SMART—specific, measurable, achievable, realistic, and bound by time. This is what the Smart monitoring and control system for greenhouse is aiming for:

- To create a smart monitoring and control system that automates the regulation of environmental conditions in a greenhouse.
- To collect and archive real-time environmental information (including temperature, humidity, soil moisture, and light levels) using a wireless sensor network.
- To develop and execute a web server interface that allows for remote access, monitoring, and manual control of the system.
- To set up a database system to record, analyze, and visualize sensor data, aiding in making informed agricultural decisions.
- To evaluate how reliable and responsive the system is when faced with varying environmental conditions.

To develop a Smart monitoring and control system for greenhouse that can handle light intensity, temperature, motion detection, and even keep track of plant health issues. It’s all about making agriculture smarter and more sustainable!

1.6. Project plan

The main goal of building a smart monitoring and control system for greenhouse is all about enhancing the plant growth environment. The dependence on utilizing the most advanced sensor and automation technologies, along with thorough analysis of historical data. As demonstrated, leveraging these tools enables the establishment of improved conditions for the flourishing of our plants. It's fascinating how all these contemporary technologies combine to aid in something as fundamental yet crucial as cultivating plants.

Step 2 – Identify Individual Objectives:

- To create a smart monitoring and control system that automates the regulation of environmental conditions in a greenhouse. A system will be created that utilizes a microcontroller (for instance, the ESP32) to gather data from sensors and activate devices like fans, water pumps, and lights. This system will be set up to automatically adjust to variations in temperature, humidity, soil moisture, and light intensity without needing manual intervention, thereby maintaining the best conditions for growth.

- To collect and archive real-time environmental information (including temperature, humidity, soil moisture, and light levels) using a wireless sensor network. A wireless network of sensors will be set up inside the greenhouse to continuously track environmental conditions. These sensors will transmit real-time data to the main controller, where the data will be stored and utilized to guide control actions or notify the user when conditions go beyond established limits.

- To develop and execute a web server interface that allows for remote access, monitoring, and manual control of the system. A web-based interface will be developed to enable users to remotely access live sensor data and manage greenhouse devices. The server will be hosted either locally or online, offering user-friendly access via any smart device, allowing users to interact with the system remotely from any location with internet access.

- To set up a database system to record, analyze, and visualize sensor data, aiding in making informed agricultural decisions. A structured database is set to be integrated for the purpose of storing all sensor readings over time. This data will facilitate the generation of visual insights, such as charts and graphs, via the web interface. Such insights will assist users in analyzing trends, optimizing crop conditions, and making informed, data-driven decisions to enhance plant growth and improve resource efficiency.

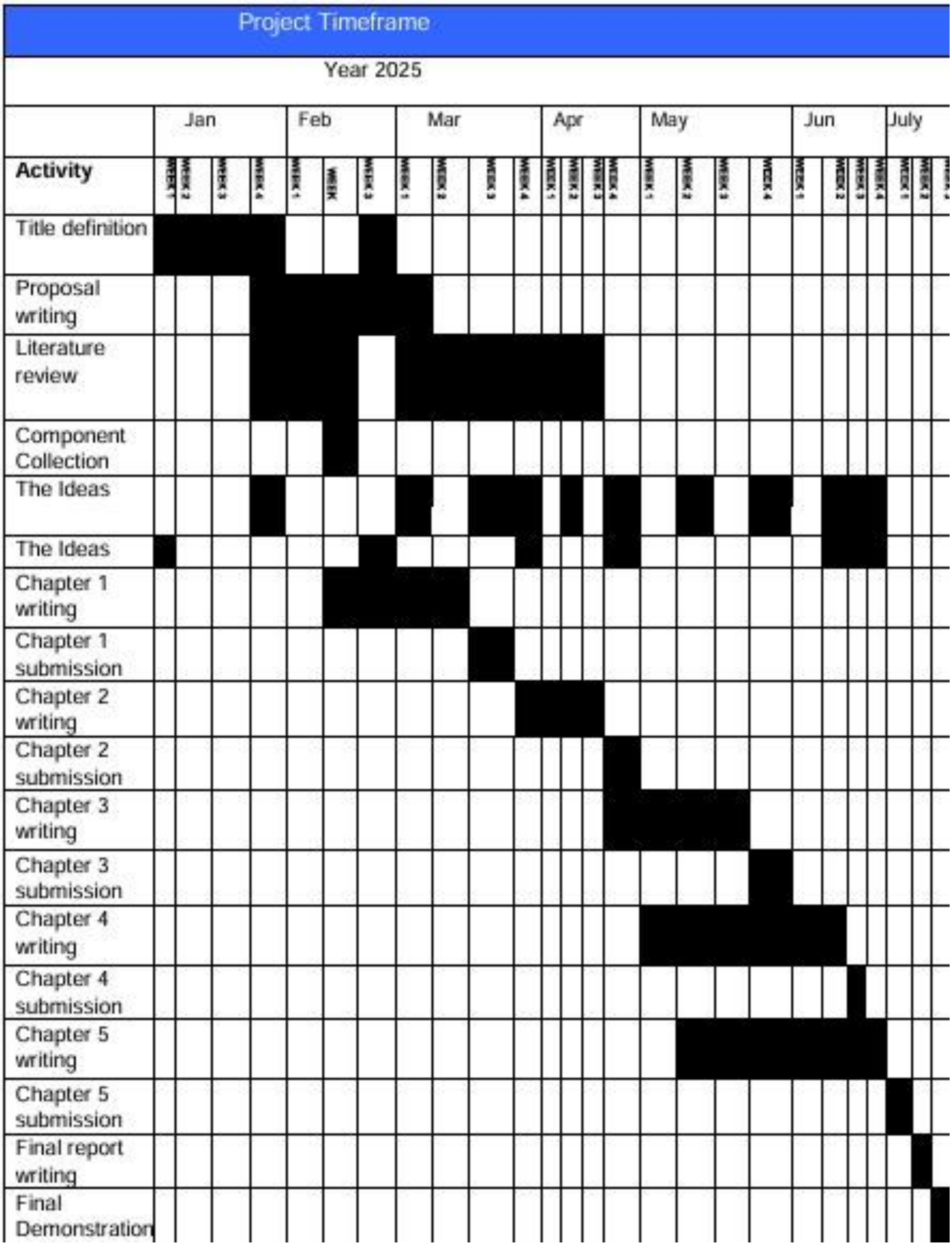
To evaluate how reliable and responsive the system is when faced with varying environmental conditions. The system's performance will be evaluated in various simulated environmental conditions to measure its responsiveness and reliability. Important metrics like reaction time, sensor accuracy, actuator response, and stability of data transmission will be collected and analyzed to identify areas for enhancement and guarantee the system's strength.

These goals will steer the development and implementation of the smart monitoring and control system for greenhouse, ensuring movement toward creating a greenhouse environment that's more efficient and sustainable. By tackling these objectives, the system is set to provide valuable analysis and practical solutions aimed at boosting crop productivity, optimizing greenhouse conditions, and promoting sustainable farming practices

1.7 Proposed budget

Description	Cost
ESP32-WROOM-32 Microcontroller	R108
Temperature sensor	R127
Humidity sensor (soil moisture sensor Module)	R34
Light Dependent Resistor (LDR) sensor(x2)	R4,79
Passive infrared sensor (PIR) module	R40
Jumper wires	R65
Breadboard	R20
Buzzer	R25
LEDs x2	R2
Total	**R425.79**

1.8 Time Schedule



1.9 Limitations

Each study needs to provide various real information and essential requirements for a smart monitoring and control system for greenhouse. The different limitations are as follows:

Technical usage requires certain technical skills to create and maintain a smart monitoring and control system. Farmers who have not previously utilized technology will find it challenging to identify or rectify problems. This system is not an accepted platform. However, it remains an unresolved issue. The uncertainty lies in what actions we should take regarding the problem.

Power supply dependency: Smart monitoring and control systems for greenhouses, which possess the ability to consider power sources for power-related issues, are essential. Providing sustainable energy can be difficult in areas where electricity is frequently cut off. The local environment presents hazardous natural conditions between electronic equipment and testing tools, including extreme cold and low-temperature levels. Sustainable maintenance and protective measures are necessary to ensure continuous safety. To ensure their long-term health, we need constant maintenance and precautionary measures.

Restricted Customisation for different crops, the system might not adjust to the needs of some crops, resulting in low efficiency in plant growth.

1.10 Conclusion

To conclude, the first important step towards enhancing the stability of a greenhouse and production is to build and implement a smart monitoring and control system in the greenhouse.

The solution aims at long-term smart control systems in greenhouses and seeks to eliminate errors and issues through the integration of advanced sensor technology, automation, and data analysis. The proposed system closely monitors and regulates the vital elements of the environment, such as soil, light, humidity, and temperature, thereby improving the conditions for plant cultivation. The project is significant as it reduces operational costs, enhances equipment performance, and supports sustainable agricultural practices.

Starting from design and system applications, the review will be completed promptly.
The configuration and evaluation of data will continue to strengthen the process. Focusing on these key goals, the project provides insightful analysis and solutions for modern greenhouse management which enhance environmental sustainability and food security. If this smart control system is successfully implemented, it will revolutionize greenhouse farming and establish new standards for production and efficiency.

Preview of the Upcoming Chapter

Various microprocessors that can be used to enhance the smart control system of the greenhouse and the necessary supporting devices to achieve the final solution will be discussed in detail in Chapter 2. To provide a comprehensive understanding of the technical aspects of the project, this chapter offers a complete overview of the technical components required for the smart monitoring and control system for greenhouse to be successfully implemented.

Chapter 2: Research Design and Methodology

2.1 Introduction

This chapter focuses on the research design and methodology for a Smart monitoring and control system for greenhouse using a microcontroller. This system is made of sensors that will monitor temperature, humidity, and light intensity, help with communication remotely, and a microcontroller that functions as the brain of a system used for data processing and direct control.

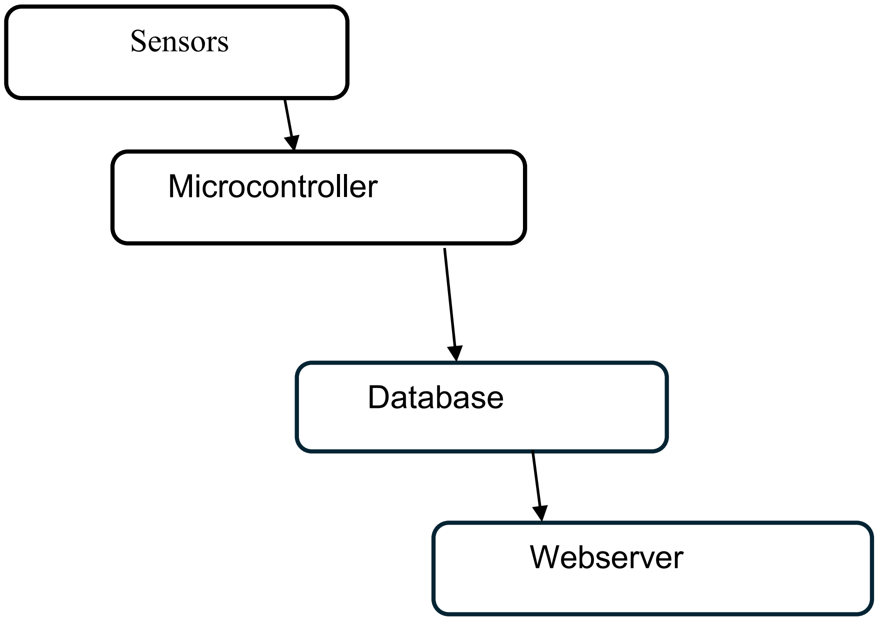


Figure 2 Block diagram

2.2 Scope of work

The Smart monitoring and control system for greenhouse is designed using an ESP32 microcontroller, integrated sensors (LDR, temperature, humidity, and infrared), Wi-Fi, a webserver, and a MySQL database. Through the automation of environmental management, the system aims to ensure conducive conditions for plant growth.

2.2.1 ESP32 Microcontroller

Figure 3 ESP32 Microcontroller

A microcontroller (MCU) is a compact integrated circuit specifically designed to manage a particular function within an embedded system. It integrates the functionalities of a central processing unit (CPU), memory, and input/output interfaces onto a single chip. Microcontrollers are extensively employed across diverse sectors, including domestic appliances, automotive systems, medical devices, and industrial control systems. There are different types of microcontrollers, such as Arduino Uno, STM32, ESP32, etc. The ESP32 microcontroller was chosen over other options because it has built-in Wi-Fi and Bluetooth features, which allow for easy connectivity in the Internet of Things (IoT) environment. It has a dual-core processor that can handle multiple tasks at once, such as checking sensors and communicating with servers. Additionally, its very low power usage improves energy efficiency, and its affordability makes it ideal for large-scale implementations (Espressif Systems, 2020). The device works well with the Arduino Integrated Development Environment (IDE) and has a wide range of library support, making it easier to prototype and connect with different sensors, actuators, and databases, which is especially beneficial for smart greenhouse systems. The ESP32 WEMOS D1 R32 Microcontroller block acts as the central department (the brain) for information processing in the system, where it processes information from all sensors and units and performs operations programmed based on input. It guarantees data processing and time-consuming decision-making and ensures the automation of environmental monitoring in a greenhouse.

2.2.2 Passive Infrared Sensor

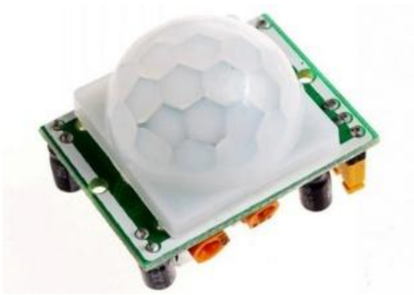


Figure 4 Passive Infrared Sensor Module

Infrared (IR) sensors are devices specifically created to detect infrared energy that is either emitted or reflected by objects. They support various functions, including motion detection, proximity sensing, and thermal imaging. There are different types of IR sensors, such as thermal IR sensors (used for temperature measurement), active IR sensors (which project infrared light and measure the reflected light to determine distance), and Passive Infrared (PIR) sensors (which identify changes in ambient infrared radiation due to movement). PIR sensors are often favored in greenhouse settings due to their energy-efficient, cost-effective, and passive nature, allowing them to sense motion like that of pests or humans without sending out signals, which boosts their reliability and minimizes false alerts. PIR sensors are energy-efficient, cost-effective, and well-suited for greenhouse applications where distinguishing between moving and stationary elements is essential (Singh et al., 2019) Their capability to differentiate between moving objects and stationary backgrounds fits well with sustainable automation objectives, making them excellent choices for security, occupancy monitoring, and energy-efficient lighting control in intelligent farming systems.

2.2.3 LDR Sensor

Figure 5: Light Dependent Resistor (LDR) Sensor

Light Dependent Resistor (LDR) sensors are classified as passive electronic components that modify their electrical resistance in reaction to fluctuations in light intensity. This property facilitates their application in a multitude of domains, including automatic lighting systems, utilization of natural daylight, and monitoring of environmental parameters. Various types of LDRs are available, such as the GL5528, which is noted for its affordability and versatility in general light sensing applications; the GL5537-1, recognized for its high sensitivity in conditions of low light; and the NORPS-12, which exhibits robust performance across a wide range of environmental conditions. LDRs are preferred due to their simplicity, cost-effectiveness, and reliability in light-sensitive applications, as they require minimal additional circuitry to function effectively. Their efficacy is particularly critical in situations demanding fundamental light detection without extensive calibration, such as in the management of shading systems within greenhouses (Patel and Mehta, 2020). The LDR block monitors light intensity in a greenhouse and transmits the data captured to the ESP32 microcontroller. It ensures that plants receive proper lighting conditions which leads to the best plant growth.

2.2.4 Humidity Sensor

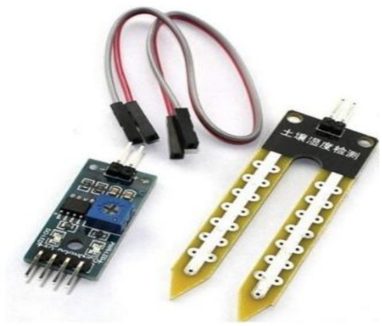


Figure 6 Humidity Sensor

Humidity sensors are devices that quantify the moisture content present in the air, referred to as relative humidity, and are integral components of climate control systems. Notable varieties include capacitive sensors, such as the DHT22 and SHTC3, alongside resistive sensors. In contrast, soil moisture sensor modules, exemplified by the FC-28 and Capacitive Soil Moisture Sensor v1.2, are specifically engineered to ascertain the water content within soil. These modules are particularly advantageous for greenhouse systems as they facilitate direct monitoring of irrigation requirements, help mitigate the risk of overwatering and optimize the hydration of plants. Unlike general humidity sensors, soil moisture sensors offer precise, localized information pertinent to root zones, thereby ensuring efficient water usage and promoting crop health. This functionality renders them essential for sustainable greenhouse practices and automated irrigation management (Gupta and Jha, 2018).

The humidity sensor block determines the soil moisture level and sends the determined data to the microcontroller. This helps in managing humidity levels and forms a conducive environment for plants.

2.2.5 Temperature Sensor

Figure 7 Temperature sensor

Temperature sensors assess thermal energy to gauge the heat level of an environment, transforming this energy into an electrical signal for the purposes of monitoring or control. Some commonly used sensors are thermocouples (known for their wide range and durability), RTDs (which provide high accuracy), thermistors (offering quick response times), and integrated circuit-based sensors such as the DS18B20. The DS18B20 is preferred due to its digital output (which minimizes noise interference), broad temperature range (-55°C to +125°C), waterproof construction (making it suitable for monitoring in soil or liquids), and 1-Wire interface that permits multiple sensors to connect to a single microcontroller pin. Its high precision (±0.5°C accuracy), minimal power usage, and straightforward integration with systems like the ESP32 make it an excellent choice for greenhouse applications that depend on dependable and scalable temperature monitoring. The temperature sensor reads the temperature of the ambient time range and sends this information to the microcontroller. It allows thorough temperature control which is essential for plants' health and growth (Maxim Integrated, 2019).

2.2.6 Webserver

Figure 8 Webserver

A webserver refers to either software or hardware designed to process HTTP requests and provide web content, such as HTML pages, APIs, or data, to clients via the internet. Among the most widely used webservers are Apache, Nginx, and Microsoft IIS. Although Python Flask is technically classified as a lightweight web framework rather than a standalone server, it is frequently utilized in conjunction with production-grade servers like Gunicorn or uWSGI for the development and deployment of web applications. Flask is favored for its simplicity, flexibility, and minimalist design, allowing for the rapid development of scalable applications equipped with features such as routing, templating, and RESTful API integration. Additionally, its compatibility with IoT systems, such as ESP32 microcontrollers, and its ease of connectivity to databases like MySQL render it particularly suitable for smart greenhouse control systems that necessitate real-time monitoring and user-friendly interfaces (Grinberg, 2018).

2.2.7 Database

Figure 9 Database

A database is an organized system designed for the efficient storage, arrangement, and retrieval of digital data. The most common types of databases are relational databases (for instance, MySQL and PostgreSQL), NoSQL databases (such as MongoDB), and embedded databases (like SQLite). MySQL is often preferred due to its open-source nature, scalability, and strong performance in managing complex queries and large datasets. It adheres to ACID (Atomicity, Consistency, Isolation, Durability) principles to ensure reliable transactions, integrates effortlessly with web frameworks such as Python Flask, and provides robust security features. Additionally, its compatibility with IoT systems (e.g., ESP32) and tools for real-time data analysis position it as an excellent choice for applications like smart greenhouses, where structured data storage and swift retrieval are essential for operational efficiency and informed decision-making (Elmasri and Navathe, 2015).

2.2.8 Buzzer

Figure 10 Buzzer

A buzzer is characterized as an electromechanical or piezoelectric device that produces audible sound signals, which encompass alarms and alerts, through vibrational mechanisms. The most common types of buzzers include piezoelectric buzzers, which operate at high frequencies while using minimal power, and electromagnetic buzzers, which utilize a coil to generate a louder sound output. The DC5-12V electromagnetic buzzer is particularly valued for its increased sound intensity, durability, and compatibility with microcontroller systems, such as the ESP32, due to its wide operating voltage range. Its sturdy design guarantees reliable operation in high noise environments, such as greenhouses, making it especially suitable for critical alerts, including pest detection or system failures, while its low power consumption meets the standards of energy-efficient automation (Kumar and Yadav, 2021).

2.2.9 LED

Figure 11 LED

An LED (Light Emitting Diode) is a semiconductor device that emits light upon the application of an electric current, and it is frequently employed for various applications including indicators, displays, and general lighting. The types of LEDs available include SMD (Surface-Mounted Device) LEDs, high-power LEDs designed specifically for illumination, and 5mm diffused LEDs. The latter are encased in a rounded, frosted lens that facilitates wide-angle light dispersion. The 5mm diffused LED is particularly preferred for its extensive viewing angle, rendering it especially suitable for status indicators, along with its robustness and compatibility with low-voltage circuits such as 3.3V to 5V microcontrollers, exemplified by the ESP32 model. The diffused lens of this LED serves to soften light intensity, mitigate glare, and promote visibility from various angles. Consequently, it is well-suited for user interfaces or visual alerts in settings such as greenhouses, where distinct and energy-efficient signaling is of utmost significance (Bhardwaj and Srivastava, 2020).

2.3Conclusion

To conclude, this chapter looked at the research design and methodology for developing a smart monitoring and control system for greenhouse using an ESP32 microcontroller and sensors (LDR, temperature, Humidity, and infrared), Wi-Fi, webserver, and MySQL database. It shows how these elements work together to monitor and control different environmental factors efficiently. By building a solid foundation for collecting, processing, and communicating data.

Chapter 3 will dwell on the hardware design of the project, focusing on the integration of the ESP32 microcontroller with the different sensors and levels integrated in the second chapter. Some ideas will be valid in this type of tool, opening the door to real-world implementation.

Chapter 3: Hardware design

3.1 Introduction

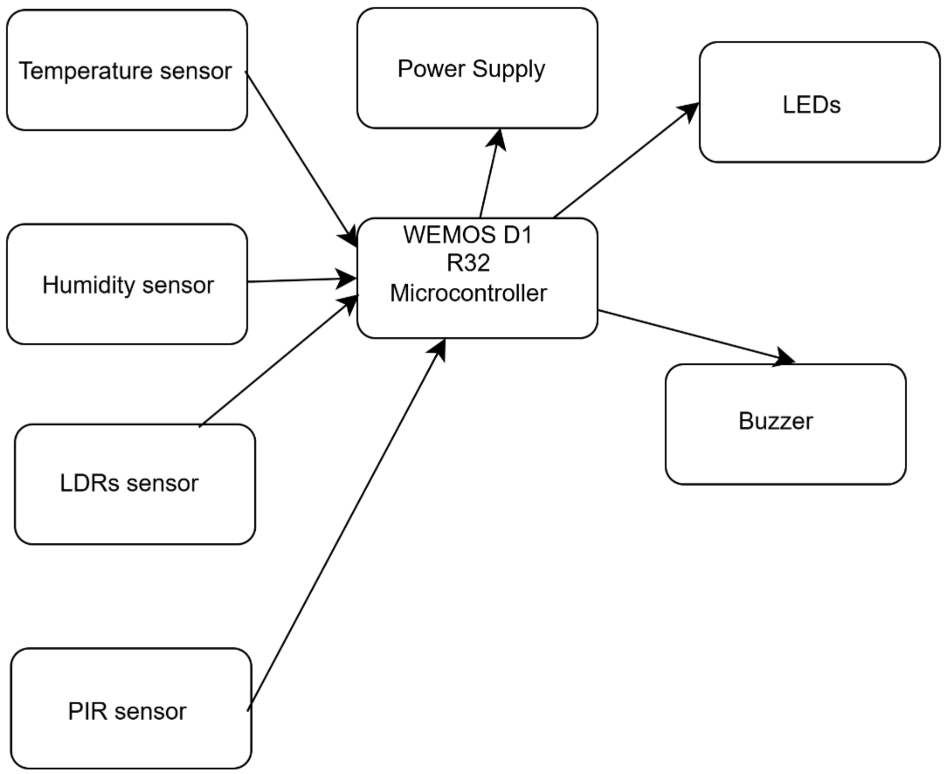


Figure 12: Hardware Block diagram

This chapter examines the hardware design of the smart control system of greenhouse, outlining the integration of important components along with the ESP32 microcontroller to facilitate effective data management and system functionality. Sensors—including LDR, temperature sensor, soil moisture, Passive Infrared Sensor (PIR), and ultrasonic sensors—along with output devices such as LEDs and buzzers, will be integrated to ensure environmental monitoring and relevant information dissemination. This ensures seamless communication between components while enhancing performance in real-world applications. This chapter provides an in-depth examination of the circuit designs, testing protocols, and validation outcomes for critical subsystems, commencing with the power supply.

3.2 Hardware design

The hardware design consists of modular components interconnected with the ESP32 microcontroller, which includes the following elements: Power Supply Circuit, Sensor Modules (such as LDR, Temperature, soil moisture, and PIR) - Actuator Control Circuits (relay for lights), Communication Module (utilizing Wi-Fi for data transmission). Each module is illustrated through a circuit diagram that aligns with the block diagram provided in Chapter 2 (Figure 1). Testing was performed to verify functionality, accuracy, and reliability.

3.2.1 Electronic Circuit Designs

3.2.1.1 Power Supply Circuit Design

A USB power supply is employed, typically delivering 5V regulated direct current, which is appropriate for powering the ESP32 microcontroller and its peripheral sensors. Additionally, a diode is included for reverse polarity protection and a capacitor is integrated to stabilize the supply. The power supply circuit delivers stable dual-voltage outputs of 3.3V and 5V, which are suitable for components such as PIR sensors and microcontrollers like the ESP32. A 5V linear regulator, exemplified by the LM7805, is utilized to reduce higher input voltages (for instance, 9V or 12V) for powering 5V PIR sensors. In a parallel manner, a 3.3V low-dropout (LDO) regulator, such as the AMS1117, ensures a safe voltage for sensitive logic circuits. Decoupling capacitors, comprising a 10µF electrolytic capacitor and a 0.1µF ceramic capacitor, play a crucial role in filtering noise, while a shared ground plane enhances overall stability. The presence of dual rails promotes compatibility between 5V sensors and 3.3V microcontrollers, where voltage dividers or level shifters effectively resolve signal discrepancies. This design prioritizes reliability, efficient heat management, and optimal power delivery, facilitating the seamless integration of actuators and sensors, while minimizing voltage fluctuations and mitigating potential damage to components in applications such as smart greenhouses.

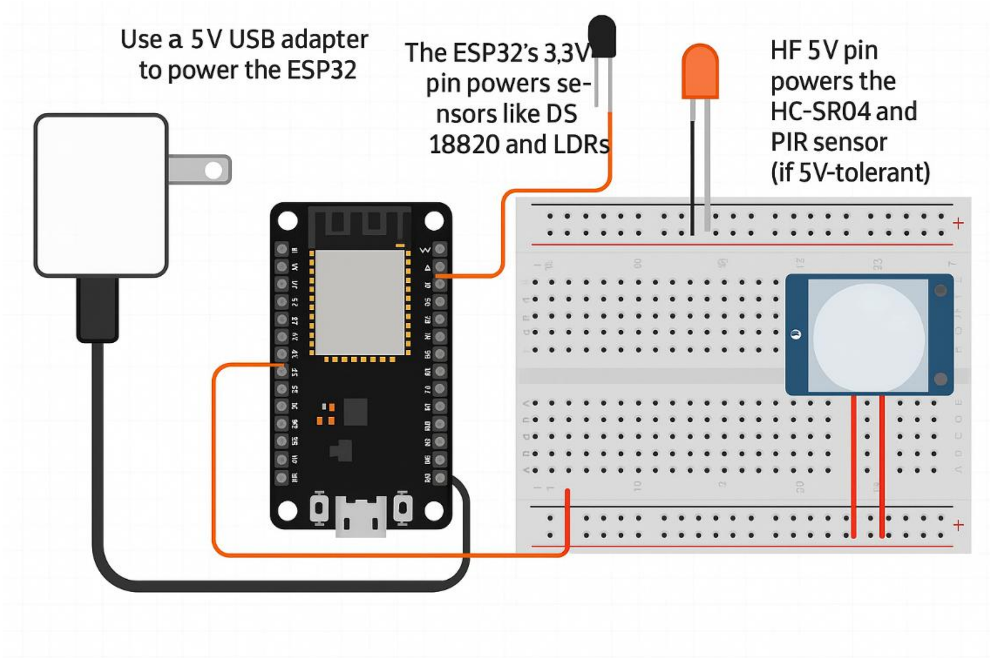


Figure 13: Power supply Circuit diagram

P=V×I

- P: Power (Watts, W)
- V: Voltage (Volts, V)
- I: Current (Amperes, A)

3.2.1.2 Relay output

The relay output circuit serves to manage high-power devices that are activated in accordance with sensor readings. Light Emitting Diodes (LEDs) are utilized to provide a visual indication of low light intensity, while a buzzer is employed to alert the user to any detected motion.

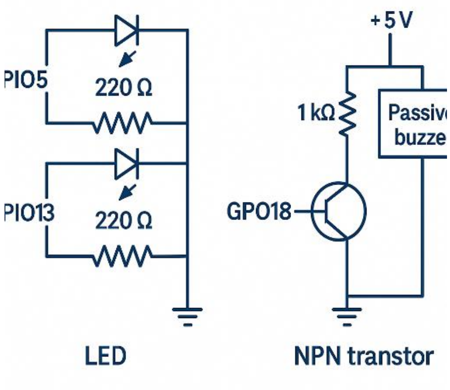


Figure 14: Relay output circuit diagram with LEDs and buzzer connection

Ohm's Law:
$R = (V_{Supply} - V_{LED})/I_{LED}$
R= Resistence
V_{LED} =Voltage of Led
I_{LED} = current of Led

```
const int led1Pin = 5;         // LED 1 control pin
const int led2Pin = 13;        // LED 2 control pin
const int buzzerPin = 19;      // Buzzer output pin
```

3.2.1.3 Sensors (Temperature, Light Dependent Resistor, Soil Moisture, Passive Infrared)

3.2.1.3.1 Temperature sensor

The temperature sensor measures the ambient temperature range and transmits this data to the microcontroller. This capability enables precise temperature regulation, which is critical for the health and growth of plants

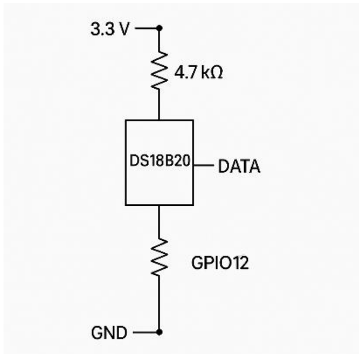


Figure 15: DS18B20 Temperature Circuit diagram

The DS18B20 temperature sensor functions by leveraging the relationship between electrical resistance and temperature to precisely measure ambient thermal conditions. It is based on the physics (Thermodynamics) principle that describes the correlation between resistance and temperature in semiconductor materials.

$$\mathbf{R(T) = R_0 * e^{B(1/T - 1/T0)}}$$

In this equation, R0 represents the resistance at the reference temperature T0, while B denotes the material constant.

```
#define ONE_WIRE_BUS 12       // OneWire bus for temperature sensor
OneWire oneWire(ONE_WIRE_BUS); // Create OneWire instance
```

DallasTemperature sensors(&oneWire); // Pass to Dallas Temperature library
const float TEMP_THRESHOLD = 35.0;   // Temperature alert threshold (°C)

3.2.1.3.2 LDR (Light Dependent Resistor):

The LDR assesses light intensity within a greenhouse and transmits the collected data to the ESP32 microcontroller. This ensures that plants are provided with optimal lighting conditions, thereby facilitating optimal growth.

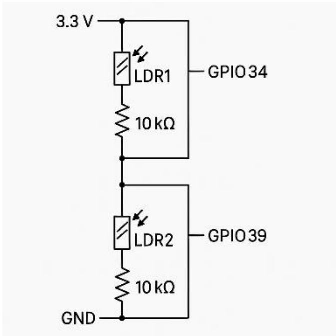


Figure 16: LDR Circuit diagram

Implement Voltage divider rule (Ohm's law) to determine resistance of LDR

$V_{out} = (R_2/ R_2+R1)$ *Vin

$V_{out} = (R_{LDR}/ R_{LDR}+10k\ \Omega)$ *3.3V

const int ldr1Pin = 36;   // Light Dependent Resistor 1 analog pin
const int ldr2Pin = 39;   // Light Dependent Resistor 2 analog pin
const int LDR1_THRESHOLD = 1800;   // Darkness threshold for LDR1 (lower = darker)
const int LDR2_THRESHOLD = 1500;   // Darkness threshold for LDR2

3.2.1.3.3 Soil Moisture Sensor:

Evaluates the moisture content present in the soil.

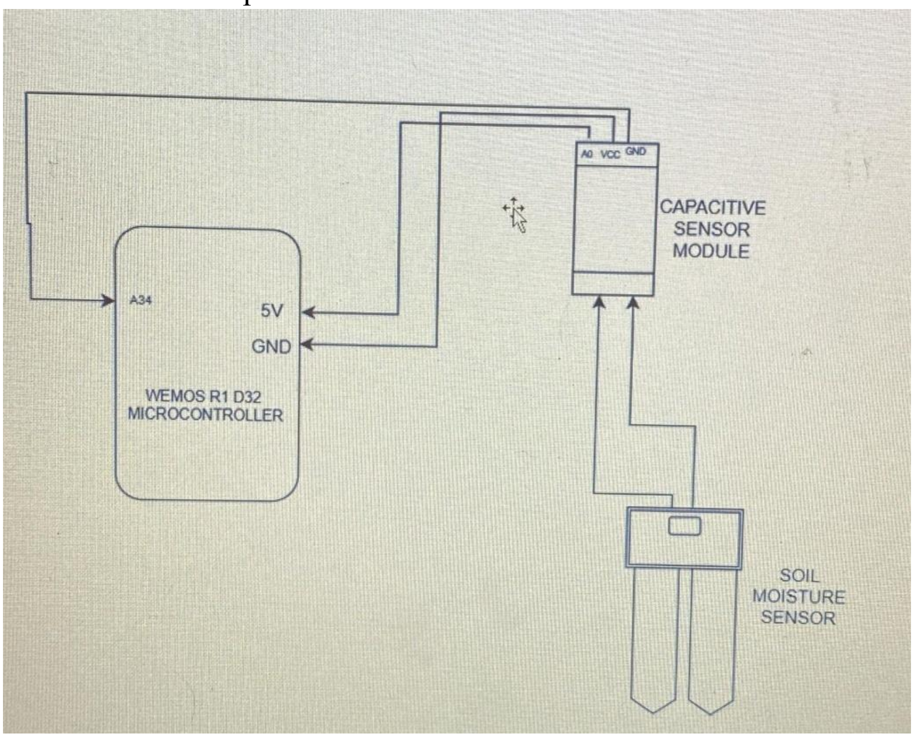


Figure 17: Soil Moisture circuit

To determine the output from the soil moisture sensor represented as ADC values with a resolution of 12 bits, the following Analog to Digital Conversion formula is used:
Moisture level = (ADC VALUE/2^n) * 100%
n = number of bits

const int soilSensorPin = 34; // Soil moisture sensor analog pin
DRY_SOIL_THRESHOLD = 3000;   // Dry soil analog reading threshold
const int WET_SOIL_THRESHOLD = 1500;   // Wet soil analog reading threshold

3.2.1.3.4 PIR Sensor:
Monitors movement for purposes of security or activity observation.

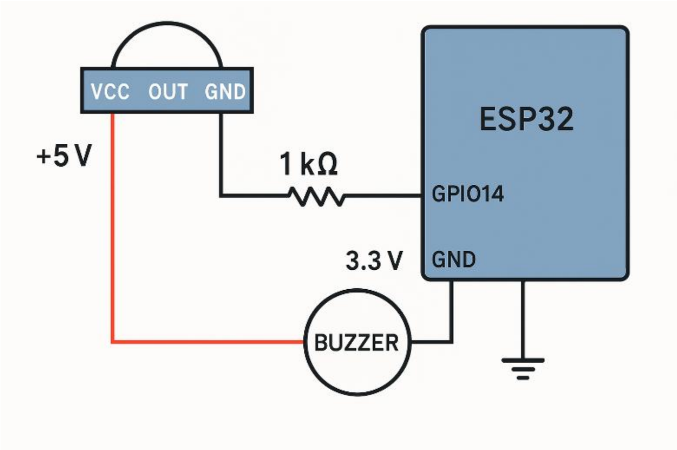


Figure 18: PIR sensor Circuit diagram

const int pirPin = 23;   // PIR motion sensor digital pin
unsigned long motionStartTime = 0;   // Timer for motion duration
bool motionActive = false;   // Current motion detection state
bool humanDetected = false;   // Confirmed human presence flag
const int minMotionDuration = 100;   // Minimum valid motion duration (ms)
const int cooldownPeriod = 100;   // Debouncing period between detections (ms)

3.2.1.4 Main circuit design

The main circuit incorporates various sensors, including passive infrared (PIR), light-dependent resistors (LDR), temperature sensors, and soil moisture sensors, in conjunction with the ESP32 microcontroller, actuators such as LEDs, and buzzers, as well as power regulation components featuring both 5V and 3.3V rails to effectively automate greenhouse operations. The ESP32 is responsible for processing the data obtained from the sensors, activating the actuators when certain predefined thresholds are met. such as turning on lights when light intensity levels drop or activating a buzzer in response to detected motion and transmitting real-time data to a web server and MySQL database using a Wi-Fi connection. The power supply circuit guarantees a stable voltage output. Furthermore, the integration of relay drivers equipped with current-limiting resistors for LEDs and buzzers facilitates the safe operation of high-power devices while also providing user feedback. The design emphasizes scalability and energy efficiency, thereby ensuring dependable environmental monitoring, optimized resource use, and remote access capabilities for the effective management of sustainable greenhouse practice.

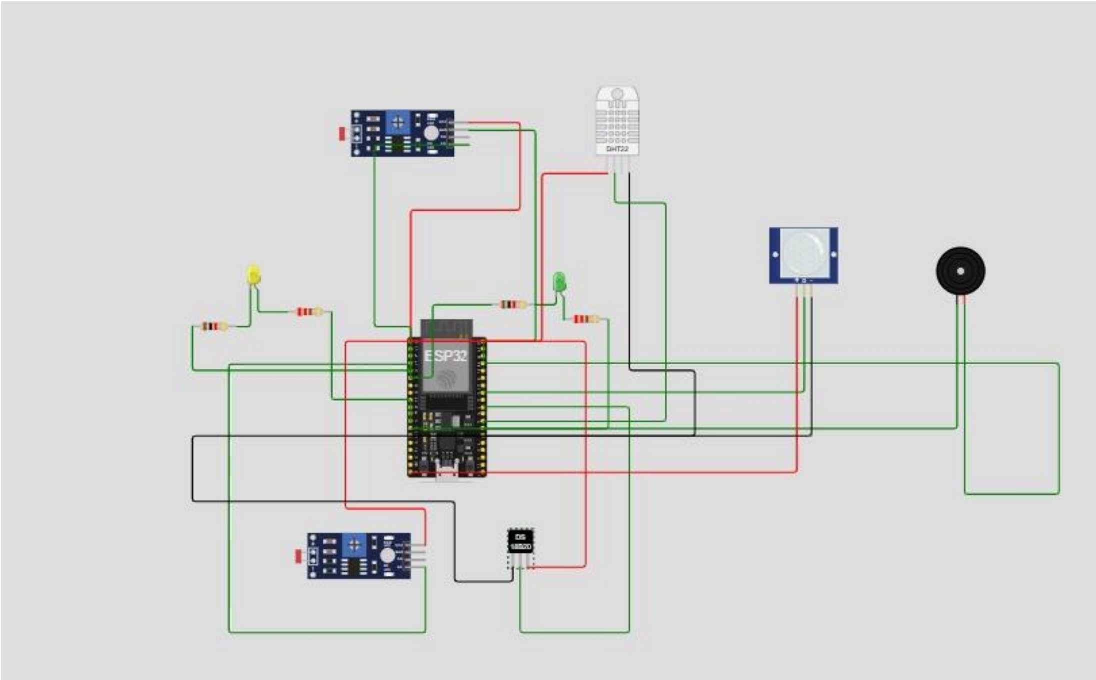


Figure 19: Main circuit diagram

3.2.1.5 Wiring

To establish a minimal, regular, and labelled connection, the ESP32 and sensors are powered through a 3.3V rail that is utilized by the DS18B20, soil moisture sensor, LDR, and LEDs, in addition to a 5V port (if necessary) for the HC-SR04 ultrasonic sensors and PIR, which functions as a common ground bus. It is advisable to employ a voltage divider to convert the 5V levels to 3.3V at the echo pin (GPIO23) and the PIR output (GPIO14) of the HC-SR04, if required, while including a 4.7KΩ pull-up resistor and an LED (220Ω) for the DS18B20. Make use of labeling wire tags. To prevent confusion, ensure that the wires are secured tightly, connect them using junctions, and mount the components onto a base. Additionally, check the connection with a multimeter and verify the sensor readings before finalizing the configuration.

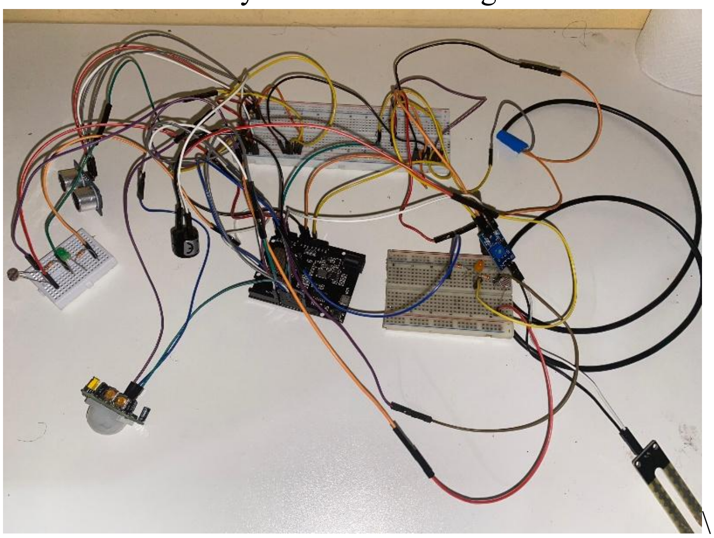


Figure 20: wiring

3.2.2 Veroboard and enclosure design

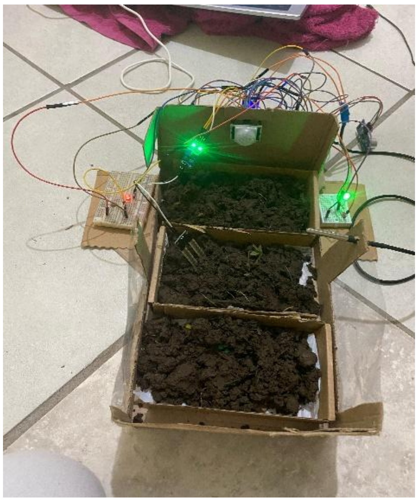


Figure 21: Illustration of a Smart monitoring and control system for greenhouse

3.3 Safety and precautions

The smart monitoring and control system for greenhouse focuses on safety and reliability by implementing electrical safeguards such as voltage isolation, fusing, reverse polarity protection, grounding, and surge suppression using diodes. It included component protection measures like current-limiting resistors, decoupling capacitors, and ESD protocols, as well as environmental resilience features including waterproof sensors and sealed enclosures. Software safety was ensured through the use of watchdog timers, encrypted Wi-Fi, data validation, and automatic MySQL backups. Operational precautions contributed to lab safety with multimeter checks, isolated high-power testing, and fire preparedness. To protect users, insulated casings, clear hazard labels, and fail-safe defaults for actuators were emphasized. Compliance with IEC/FCC standards along with thorough documentation such as schematics and manuals ensured adherence to safety protocols, allowing for strong, energy-efficient automation while reducing risks related to electrical, data, and environmental issues.

3.4 Conclusion

This chapter presents an overview of the hardware design for the Smart Greenhouse Control System, highlighting the principles of modularity, reliability, and ease of troubleshooting. Rigorous testing was conducted to validate each sensor and actuator, thereby ensuring accurate environmental monitoring and responsive control. The designs for wiring and enclosures are characterized by a focus on minimalism and accessibility, in accordance with sustainable engineering practices. Chapter 5 will explore the software architecture, encompassing sensor data processing, server communication, and user interface design**.**

Chapter 4: Software design of Smart monitoring and control system for greenhouse

4.1 Introduction

Figure 22: Software block diagram

This chapter provides a comprehensive overview of the foundation of the Smart monitoring and control system for greenhouse that is consistent with the design of the equipment presented in Chapter 4. The software is designed to automate sensor data collection, real-time decision-making, and IoT connectivity, with the aim of automating control systems in all scenarios. To enhance the Arduino IDE platform, the code effectively utilizes the power of the ESP32 to process various inputs - digital, analog, and sensor-specific data - and to create more efficient outputs such as providing controllers and notifications, in addition to maintaining regular communication with the server. Furthermore, this chapter provides a detailed description of the methods included in Chapter 3, which shows how the software converts raw data into a usable formula, which helps in the sustainable management of the green environment**.**

4.2 Software designs

The software adheres to a structured workflow designed to interpret sensor inputs, implement thresholds, and activate outputs. Presented below is the flow diagram of the system:

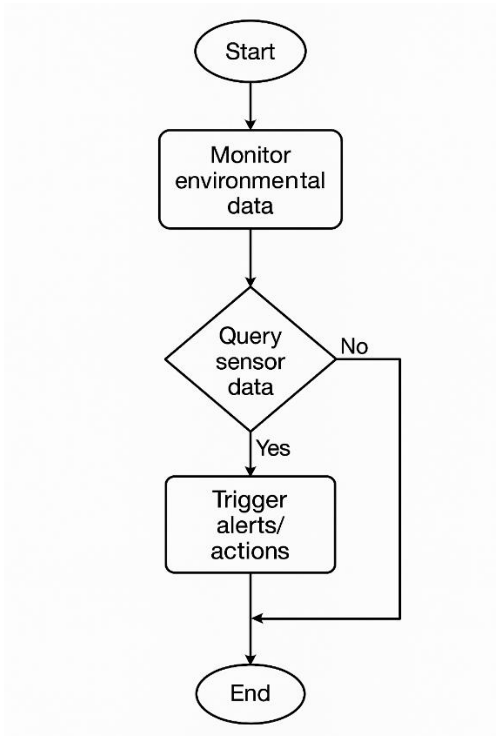


Figure 23: An activity diagram.

4.2.2 Arduino IDE Code

The Arduino IDE code pertains to the programming created and uploaded via the Arduino Integrated Development Environment, designed to control microcontroller boards such as the ESP32 Wemos D1 R32. Within a smart greenhouse monitoring and control system, this code processes data obtained from various sensors, including Light Dependent Resistors (LDRs), temperature sensors, humidity sensors, and soil moisture sensors. It subsequently manages outputs like LEDs and buzzers in accordance with the data collected. Furthermore, the Arduino code can facilitate data transmission or reception through serial communication or Wi-Fi modules, thereby allowing interaction with external systems, such as MySQL databases. In essence, the Arduino IDE code serves as the central processing unit of the system, facilitating automatic monitoring and enabling real-time responses to fluctuations in greenhouse conditions.

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <ArduinoJson.h>

// Wi-Fi credentials for network connection
const char* ssid = "DUNANIPRINCE1067";  // Wi-Fi SSID
const char* password = "@Chauke1234";   // Wi-Fi password

// Server URL to send sensor data
const char* serverUrl = "http://10.81.192.33:5000/api/sensor/batch";

// Sensor Pins
#define ONE_WIRE_BUS 12
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

const int soilSensorPin = 34;   // Pin for soil moisture sensor (FC-28)
const int ldr1Pin = 36;         // Pin for first LDR sensor
const int ldr2Pin = 39;         // Pin for second LDR sensor
const int pirPin = 23;          // Pin for PIR motion sensor
const int buzzerPin = 19;       // Pin for buzzer alarm
const int led1Pin = 5;          // Pin for LED1 (LDR1 status indicator)
const int led2Pin = 13;         // Pin for LED2 (LDR2 status indicator)
```

Figure24: Arduino Code

4.2.3 FRONT-END

The front-end represents the user interface (UI) that enables farmers or operators to engage with the smart monitoring and control system for greenhouse through a web-based dashboard. Developed utilizing frameworks such as Python Flask for the backend and Bootstrap/HTML/CSS for the frontend, it presents real-time sensor data regarding temperature, humidity, light, and soil moisture, as well as historical trends and system status. Users are empowered to monitor conditions remotely, establish environmental thresholds, and exert manual control over actuators, such as lights, fans, and irrigation systems. The front-end maintains communication with the ESP32 microcontroller and MySQL database to retrieve live data, visualize this information through charts and graphs, and issue commands for necessary adjustments, thereby facilitating intuitive, accessible, and responsive management of greenhouse operations from any device equipped with web capabilities.

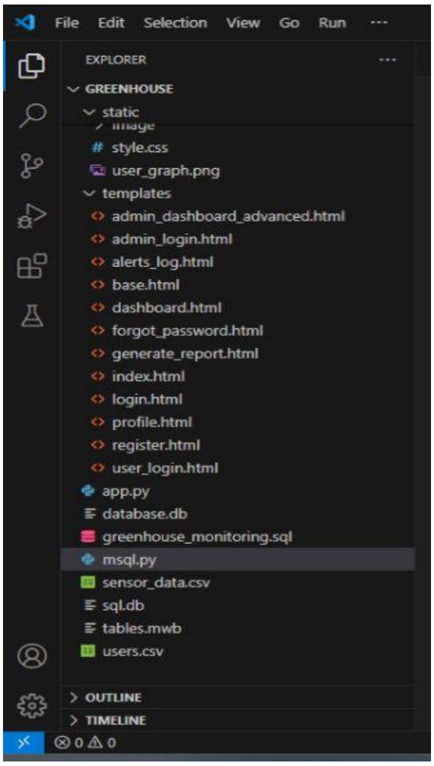


Figure 25: Front-end

4.2.4 Backend

The backend is the server-side framework responsible for processing data, managing logic, and facilitating communication between hardware and the user interface. Developed using Python Flask, it collects real-time sensor data (such as temperature and humidity) from the ESP32 microcontroller via Wi-Fi, validates and stores this information in a MySQL database, and executes automated control logic (for instance, activating a buzzer upon motion detection). Additionally, it processes user requests from the frontend (for example, modifying settings and retrieving historical data) through RESTful APIs, thereby ensuring secure data transmission and seamless integration with actuators (like lighting systems). By orchestrating the flow of data, storage, and decision-making, the backend guarantees the reliable, scalable, and secure operation of the greenhouse automation system.

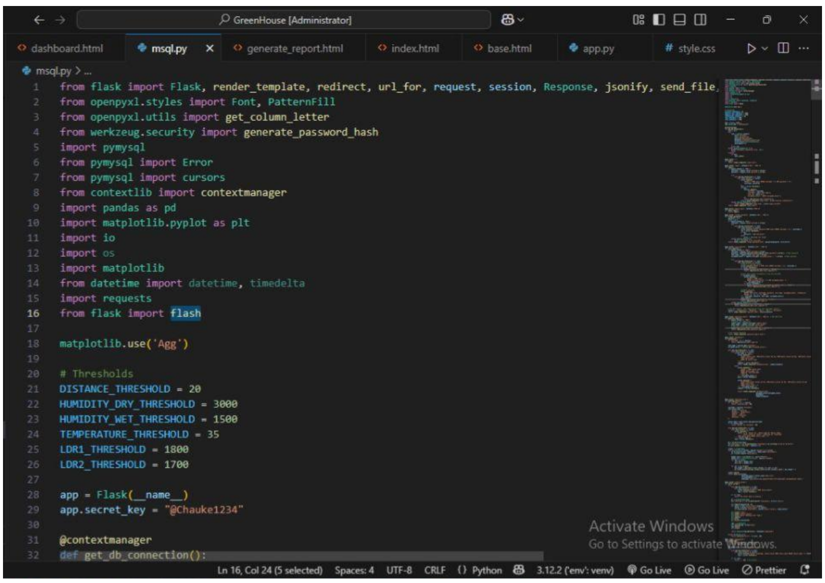
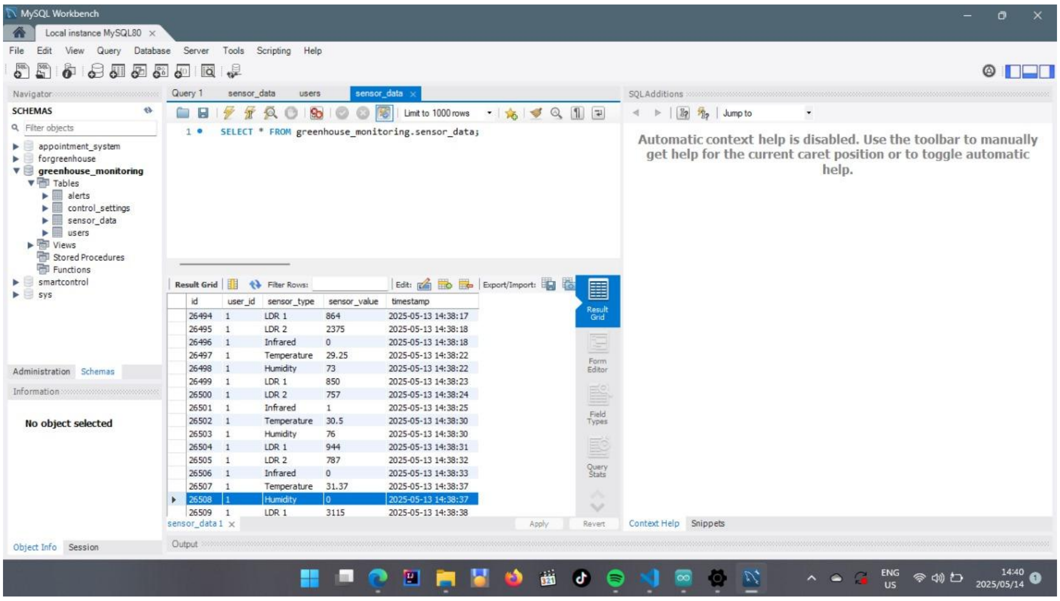


Figure26: Backend

4.2.5 Database structure

A database, such as MySQL, serves as a structured system designed for the efficient storage, management, and retrieval of data. Within the context of a smart greenhouse monitoring and control system, MySQL is utilized to store environmental data, including temperature, humidity, light intensity, and soil moisture, which is gathered from sensors. Additionally, it accommodates control data for devices, such as buzzers and lights. SQL queries, including `SELECT`, `INSERT`, `UPDATE`, and `DELETE`, are employed to obtain sensor readings, log new data, modify control settings, and manage records. MySQL facilitates real-time data handling, bolsters decision-making processes, and ensures the efficiency of the automation system by providing reliable data storage and access.



Fugure27: Database structure

4.2.3 Sequence diagram

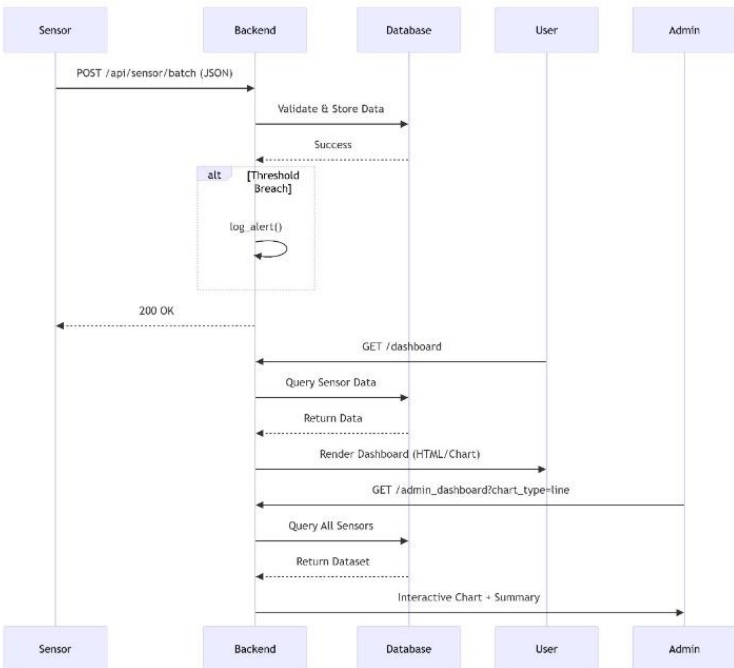


Figure 28: Sequence diagram

4.3 Conclusion

This chapter provided a comprehensive overview of the software design, incorporating a flow diagram, the configuration of the Arduino Integrated Development Environment (IDE), and annotated code. The implemented logic efficiently integrates various input types to effectively manage outputs. The subsequent chapter will discuss the problems encountered when designing a smart monitoring and control system for greenhouse and how the problems were addressed.

Chapter 5: Results and Discussion

5.1 Introduction

This chapter outlines the results and conclusions of the smart greenhouse project, emphasizing the difficulties encountered during the construction phase. Various challenges arose, such as inaccurate sensors, database communication issues, and hardware malfunctions. Solutions included recalibrating the sensors, modifying the code, and upgrading the hardware. Although the system operated as expected, there were still some drawbacks, including slow data updates and restricted scalability. Looking back on the experience, enhancements like implementing more sophisticated components and perfecting the system design could improve subsequent iterations.

5.2 Overall sensors graph display

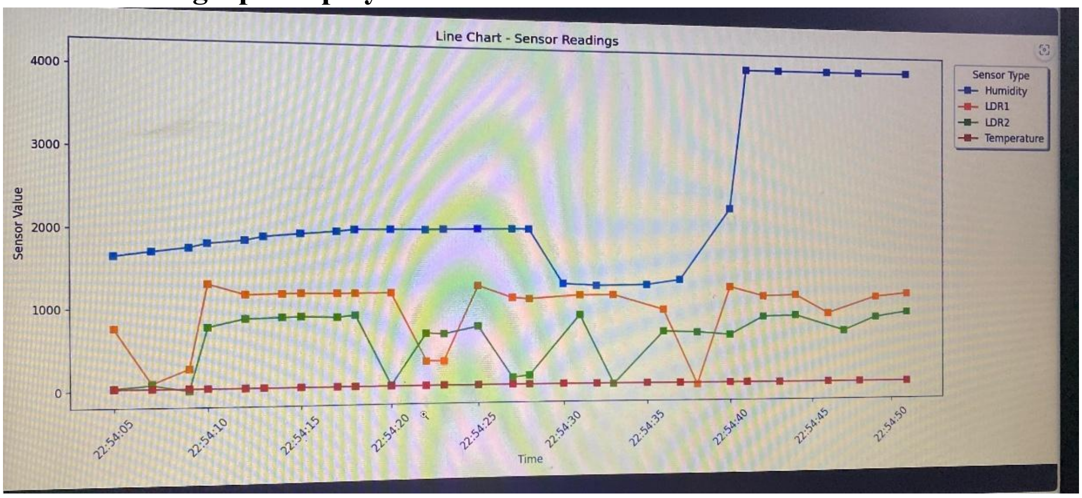


Figure 29: Overall sensor graph

5.2.1 LDR sensor graph display

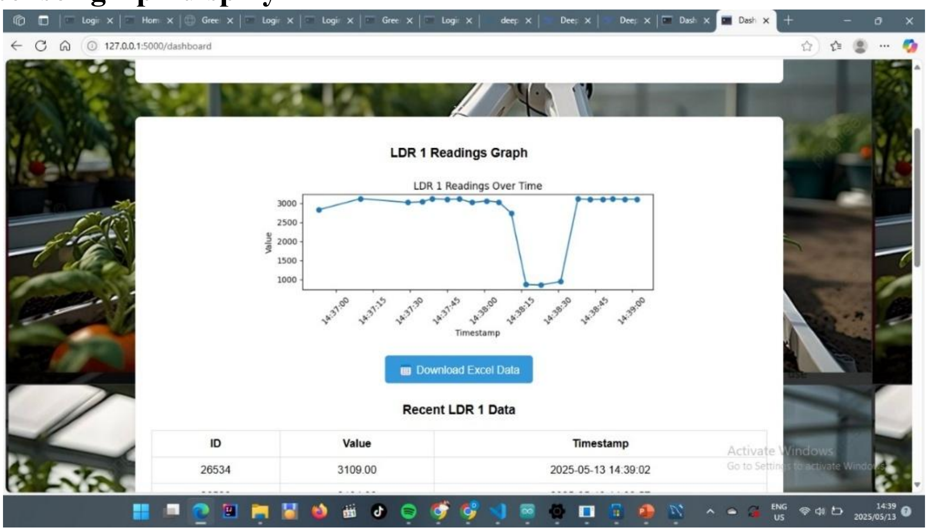
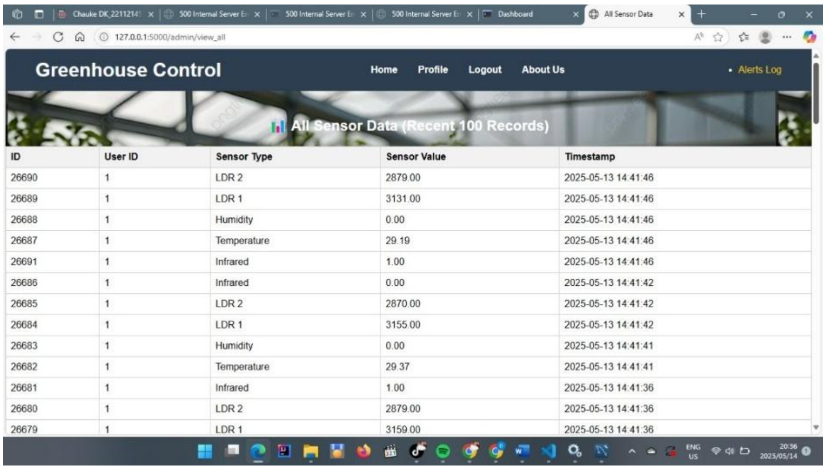


Figure 30: LDR sensor graph

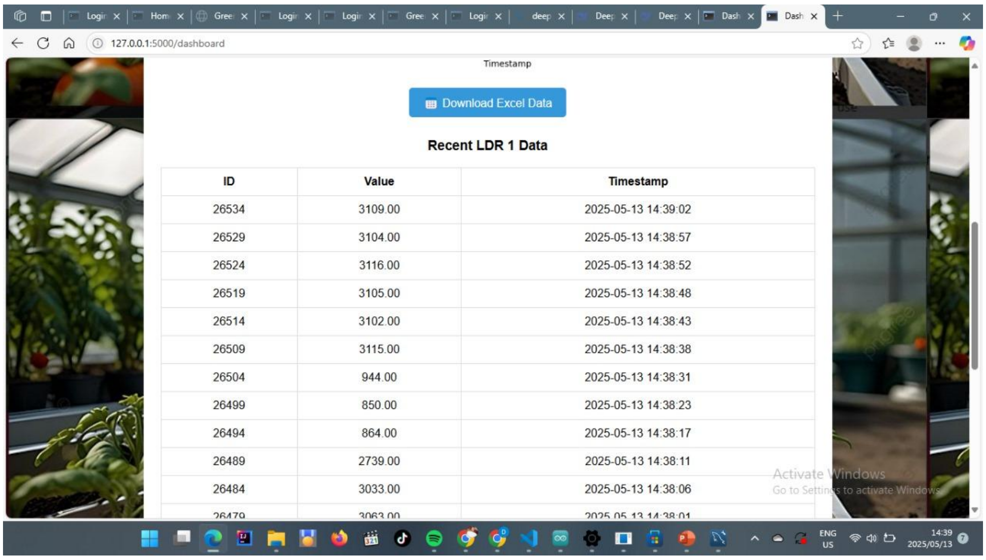
5.3 Table of all sensors on database



ID	User ID	Sensor Type	Sensor Value	Timestamp
26690	1	LDR 2	2879.00	2025-05-13 14:41:46
26689	1	LDR 1	3131.00	2025-05-13 14:41:46
26688	1	Humidity	0.00	2025-05-13 14:41:46
26687	1	Temperature	29.19	2025-05-13 14:41:46
26691	1	Infrared	1.00	2025-05-13 14:41:46
26686	1	Infrared	0.00	2025-05-13 14:41:42
26685	1	LDR 2	2870.00	2025-05-13 14:41:42
26684	1	LDR 1	3155.00	2025-05-13 14:41:42
26683	1	Humidity	0.00	2025-05-13 14:41:41
26682	1	Temperature	29.37	2025-05-13 14:41:41
26681	1	Infrared	1.00	2025-05-13 14:41:36
26680	1	LDR 2	2879.00	2025-05-13 14:41:36
26679	1	LDR 1	3159.00	2025-05-13 14:41:36

Figure 31: Table of all sensors

5.3.1 Table of LDR sensor



ID	Value	Timestamp
26534	3109.00	2025-05-13 14:39:02
26529	3104.00	2025-05-13 14:38:57
26524	3116.00	2025-05-13 14:38:52
26519	3105.00	2025-05-13 14:38:48
26514	3102.00	2025-05-13 14:38:43
26509	3115.00	2025-05-13 14:38:38
26504	944.00	2025-05-13 14:38:31
26499	850.00	2025-05-13 14:38:23
26494	864.00	2025-05-13 14:38:17
26489	2739.00	2025-05-13 14:38:11
26484	3033.00	2025-05-13 14:38:06

Figure 32: Table of LDR sensor

Chapter 6: Conclusions and Recommendations

6.1 Conclusions

The advancement of a smart monitoring and control system for the greenhouse has effectively illustrated the capacity to automate environmental observation and manage devices such as LEDs and a buzzer. This project exemplifies compliance, encompassing problem-solving and sustainability. Team collaboration was essential to the development of the entire project, and ethical practices were prioritized in the management of data. Throughout the construction phase, various challenges were encountered, including inaccuracies in sensor readings, an unstable power supply, and difficulties in maintaining consistent communication between the Arduino and the MySQL database. The challenges encountered were mitigated through techniques such as sensor recalibration, enhancements to power components, and the refinement of communication code. Nonetheless, certain limitations persisted, including delays in data transmission, a restricted wireless range, and the absence of an intuitive user interface. Should this project be revisited in the future, the incorporation of more sophisticated microcontrollers (for example, ESP32), the improvement of network reliability, and the development of a mobile application would markedly enhance both functionality and the user experience.

7.2 Recommendations

To enhance future developments, it is advisable to utilize sensors of higher quality that offer improved accuracy and durability for extended monitoring. The adoption of wireless communication protocols, such as MQTT or LoRa, may enhance the efficiency and range of data transmission. Furthermore, incorporating solar energy could bolster the sustainability of the system in remote greenhouse settings. Creating a web or mobile dashboard would also enhance user-friendliness, facilitating more effective remote control and data visualization. Subsequent research could investigate the application of machine learning algorithms for predictive control, leveraging historical data to make the system more intelligent and responsive to fluctuating environmental conditions.

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Appendix

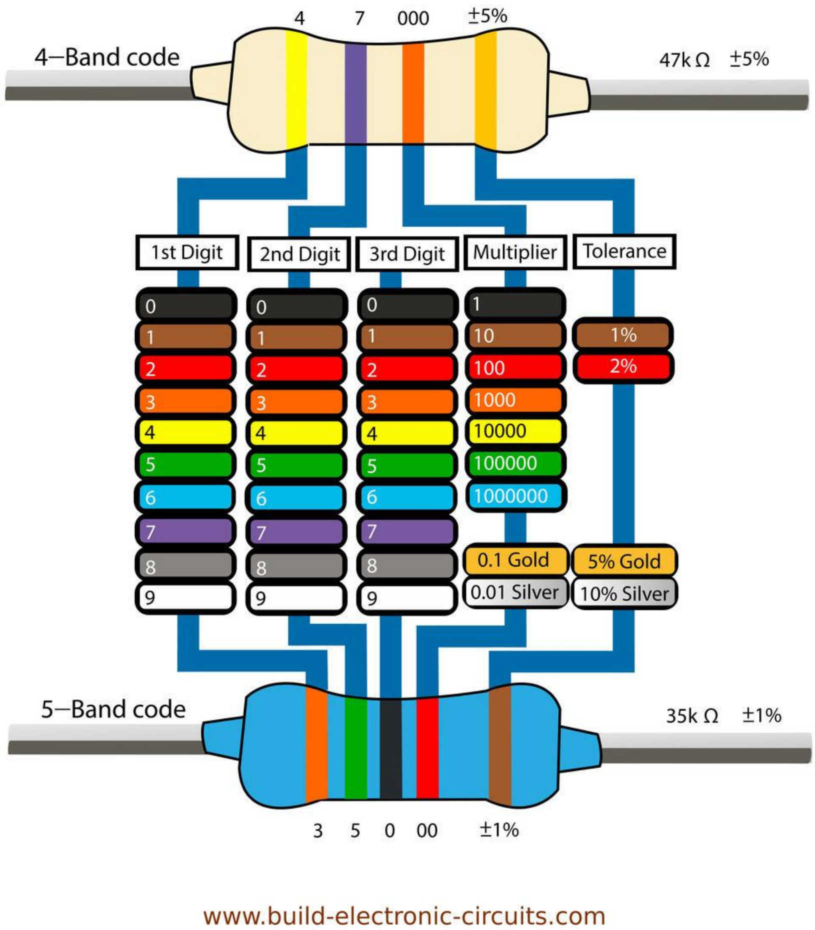

Figure 33: Resistor colour coding appendix